



Hantro SW OMX

Release Notes

Nov. 2024

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1 General Information

This guide provides an overview of the OMX SW package, including testbench compiling, for customer to kick off the package quickly.

If need more details, please refer to the documents listed in Chapter 2 "Documents List".

Release Date: Nov.07.2024

Release Package Type:<CtrlSW>

Matched VCE package:

Verisilicon_SW_VPU_Encoder_CtrlSW_v2.4.123_CL981740P_20240923.tgz.gpg

2 Documents List

```
|—— doc
|   |—— Hantro.Video.Encoder.OpenMAX.IL.API.Specifications-v0.80-A-20231205.pdf
|   |—— Hantro_SW_VC9000E_OMX-addon_CL1014875_20241107.tgz
1 directory, 2 files
```

3 Package Structure

3.1 The main structure of the OMX_Ctrlsw_Src package

Hantro_SW_VC9000E_OMX-addon_CL1014875_20241107.tgz. It's the main structure for omx ctrlsw

```
OMX_CtrlSW_20241107
|—— openmax_il
```

openmax_il the code of openmax

4 Third-party source code reference

libomxil-bellagio-0.9.3

You can get the source code from <https://omxil.sourceforge.net/downloads.html>, and keep libomxil-bellagio-0.9.3 and openmax_il in same path.

5 How to Build and install OMX component

Please untar the vce ctrlsw and kernel release package before doing vce compile, more deatils please refer to vce releasenotes.

5.1 Build and install OMX component with vce in FPGA env

```
# to generate vce ctrls libh2enc.a
$> cd [VC9000E_path]/software/linux_reference
$> make clean pci
# to build and install OMX component
$> cd OMX_CtrlSW_20241107/openmax_il/source/encoder
$> export ENC_PATH=[VC9000E code path]
$> make clean && make pci ENCODER_API_VERSION=vc9000e BUILD_OPTION="-DVCE_2_4"
$> make install
```

Then libOMX.hantro.H2.image.encoder.so & libOMX.hantro.H2.video.encoder.so will be generated in current folder

6 How to Compile test binary in FPGA ENV

6.1 Build test binary with vce

```
$> cd OMX_CtrlSW_20241107/openmax_il/source/testclient
$> make clean x86_linux_pci BIN=vce ENCODER_API_VERSION=vc9000e
Then the omxenctest will be generated in current folder
```

7 Example test commands

Here are some sample command lines for customer reference. More details about test option please run `./omxenctest`.

#HEVC:

```
./omxenctest -O hevc -s 38016 -i metro2_25fps_qcif.yuv -o stream.hevc -w 176 -h 144
-f 30 -j 30 -d -R 0 -B 384000 -q 25 -l 0 -r 0 -x 176 -y 144 -X 0 -Y 0 -C
variable -L 120 -a 0 -b 30 -n 149 -t 0 -A 1 0 -Q 1 0 --cpbSize 0 --gopSize 1
```

#H264:

```
./omxenctest -O avc -s 608256 -i metro_25fps_cif.yuv -o stream.h264 -w 352 -h 288 -f
25 -j 25 -d -R 1 -B 512000 -q 25 -l 0 -r 0 -x 352 -y 288 -X 0 -Y 0 -C
variable -L 120 -a 0 -b 30 -n 0 -t 0 -A 1 0 -Q 1 0 --cpbSize 0 --gopSize 1
```

#JPEG:

```
./omxenctest -O jpeg -s 38016 -i random.yuv -o ./stream.jpg -q 8 -a 0 -b 1 -w 96 -
h 96 -x 96 -y 96 -X 0 -Y 0 -l 0 -r 0
```

8 Supplementary Notes

1. There are some issues with the adaptation between OMX and MMU. Please temporarily disable MMU when customer test OMX on FPGA, and VSI will fix it as soon as possible.
2. When test jpeg cases on FPGA, VSI found a few cases will get segment fault for buffer free, VSI will fix it as soon as possible.